MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU–UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

MG4SECBCA201 – PROBABILITY DISTRIBUTIONS AND STATISTICAL INFERENCE

Time: 2 Hours Max. Marks : 70

PART A – Short Answer Questions

(Answer any five questions. Each question carries 2 marks)

1. Evaluate k if f(x) = k, x = 1,2,3,4,5,6 and 0 otherwise is a probability mass function. [A][CO1]

2. Define Bernoulli distribution. [R][CO2]

3. State the conditions under which Binomial distribution tends to Poisson distribution. [U][CO2]

4. What is the value of Z when the area under the normal curve is 0.5? [R][CO2]

5. State any two uses of F distribution. [U][CO3]

6. Write down the chi-square statistic for a 2×2 contingency table. [R][CO4]

7. State the relationship between level of significance and Type I error. [U][CO4]

8. Write down the test statistic for testing equality of means when population standard deviations are known. [R][CO4]

[5×2=10]

PART B – Short Essay Questions

(Answer any five questions. Each question carries 6 marks)

9. A random variable X has the following probability function:

X:	0	1	2	otherwise
f(x):	k	2k	3k	0

Find k and P(X < 2). [A][CO1]

10. Find the mean and variance of the distribution:

X:	3	8	12	
f(x):	1/3	1/2	1/6	[A][CO1]

11. If a random variable X follows a Poisson distribution such that P(X = 1) = P(X = 2), find P(X = 0). [A][CO2]

12. A coin is tossed six times. Find the probability of obtaining five or more heads. [A][CO2]

13. Define Poisson distribution and explain its important characteristics. [U][CO2]

14. The height of school children is normally distributed with mean 54 inches and standard deviation 12 inches. Find the proportion of students having height between 46 and 56 inches. [A][CO2]

15. Explain the properties of the Normal distribution. [U][CO2]

16. Explain the chi-square test as a goodness-of-fit test. [U][CO4]

[5x6=30]

PART C – Essay Questions

(Answer any two questions. Each question carries 15 marks)

17. Define Normal distribution. The distribution of marks obtained by a group of students is normal in which 7% are under 43 and 89% are under 63. Find the mean and standard deviation. [A][CO2]

18. Explain commonly used sampling distributions and their interrelationships. [An][CO3]

19. The mean life of 100 fluorescent light tubes is 1570 hours with standard deviation 120 hours. The company claims that the mean life is 1600 hours. Test the claim at 5% level of significance. [A][CO4]

20. From the following data, use chi-square test and conclude whether inoculation is effective in preventing tuberculosis. [A][CO4]

	Attacked	Not attacked
Inoculated	31	469
Non inoculated	185	1315

[2x15=30]